

# Northlake first implementation: Student Center electricity

PREPARED BY Energy Hippo implementation team

Status: **packet and source samples prepared; UI setup and installed acceptance not run.**

Use the [customer packet](#), [Stage 1B definitions](#), and [gateway coverage](#). This checklist starts a real implementation and records its evidence. It is not a database seed or an import of EEM internal tables.

## 1. Record the starting instance

- Record application/build version, database name, instance name and packet version.
- Confirm the supported product installation, setup login and intended worker are available.
- Keep the current setup account for this milestone. User/permission scenarios follow later.
- Copy `northlake-instance-mapping.template.csv` to the run's evidence directory. The generated template is overwritten when the packet is regenerated; never enter actual IDs or acceptance results into the template itself.
- Capture the actual company, hierarchy, point-type, unit and timezone choices from the UI. Numeric IDs in older scenario YAML are illustrative and must not be inserted into EEMSuite or assumed to match the installed instance.

## 2. Enter the customer foundation

Use Database Administrator's company and Nodes screens. Create only missing objects after checking the existing hierarchy.

Every company uses the same levels, listed below. A building exists once, in the company that owns its site. The Northlake Thermal Plant company contains only the Central Plant; chilled-water and steam meters at served buildings are created under those buildings in Northlake University.

| Object                       | Intended value                                                         |
|------------------------------|------------------------------------------------------------------------|
| Company                      | Northlake University                                                   |
| Hierarchy levels             | Site, Building, Meter, Point                                           |
| Hierarchy                    | System / Northlake University / Northlake Main Campus / Student Center |
| Building area                | 95,000 sq ft; address 160 Synthetic Campus Drive, Sacramento           |
| Timezone                     | Sacramento / America/Los_Angeles equivalent in the installed lookup    |
| Provider                     | Valley Electric District                                               |
| Billing account              | SYN-VED-A-0010004                                                      |
| Contractual tariff reference | VED-TOU-GS for 2024; VED-TOU-GS-FY25 for 2025                          |

| Object       | Intended value                                  |
|--------------|-------------------------------------------------|
| Meter        | Student Center Electric Main; SYN-VED-M-0040004 |
| Energy point | Delivered kWh 15m; Analog; kWh; 15 minutes      |
| Demand point | Demand kW 15m; Analog; kW; 15 minutes           |

Create data/providers/accounts through their normal UI. Full determinant construction in Rate Modeler and bill calculations are later acceptance work; they are not required for raw interval ingestion. Company calendars are required before the later bill import milestone.

The billing account needs more than its number and provider: its service address, the provider's remit address, the account representative, the bill entry and service types, the invoice template and its default GL accounts. Every account's values are listed in [the billing account setup](#); the GL accounts and the account-scope validation tests wait for their Phase 3 prerequisites.

Meters and points have the same kind of detail: provider meter number, multiplier and bill cycle on a meter, and reading type, measure type, unit, archive interval and index on a point. Each one is in [the meter and point setup](#). Enter only the points it marks Enter: EEM creates a usage and a rate-of-change point behind every register point, and the degree-day points behind a temperature index.

Record generated IDs and screenshots/exports after saving. Check that the point list contains both measurements under the correct meter. A missing required UI path is a product finding; do not fill the gap with SQL.

### 3. Configure the AcquiSuite source

Use the Gateway Configuration workspace in ConfigurationTool or the gateway workspace opened from Database Administrator. Point identities are created in Nodes; the gateway screen associates those points with a logger/device.

| Setting                                    | Value                                                         |
|--------------------------------------------|---------------------------------------------------------------|
| Customer data source                       | Campus AcquiSuite loggers                                     |
| Gateway type                               | ACQUISUITE                                                    |
| Gateway name                               | Northlake Student Center AcquiSuite                           |
| Gateway node                               | ASQ-VED-STUDENT                                               |
| Node Serial Number                         | ASQVED01                                                      |
| DeviceID on both points                    | STUDENT                                                       |
| Energy Row Number                          | 1                                                             |
| Demand Row Number                          | 2                                                             |
| RowOffset                                  | 0                                                             |
| Input pattern                              | *.log                                                         |
| Input / working / backup / error locations | Separate writable directories belonging to this test instance |

Use actual point and node IDs captured from the UI. Enable the relevant gateway object through normal instance configuration and create the supported workflow and task/schedule through the UI. Ensure the

workflow includes interval publication and that the appropriate worker is processing it. The sample is historical, so confirm the normal backfill path is active. A gateway producing an intermediate file or a successful task log alone does not prove stored data.

The format is grounded in `AcquiSuiteGatewayComponent`: filename tokens match serial/device, column zero is UTC, the configured row number plus `RowOffset` selects the value column, and the installed timezone determines local time. No header line is permitted; an unparseable timestamp ends that file's read loop.

## 4. Execute and accept

Take a configured checkpoint before introducing interval data. Use disposable restores of that checkpoint for independent cases. Copy a source sample into the configured input directory through the normal file-delivery path, then run the task from the UI. Retain the original packet files outside the watched folder.

| Case          | Input / action                                    | Required visible result                                                                           |
|---------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Normal day    | Deliver complete/ASQVED01_STUDENT.log             | 96 readings per point, correct local/UTC times, 2,020 kWh, 120 kW peak                            |
| Replay        | Deliver that same complete file again             | Still 96 unique timestamps per point and 2,020 kWh; capture the actual duplicate/replay treatment |
| Missing block | Fresh configured checkpoint; deliver the gap file | 88 readings per point, 1,780 kWh, missing 12:00–13:45 PST                                         |
| Backfill      | Deliver the backfill file after the gap           | 96 readings per point, gap closed, 2,020 kWh and 120 kW peak                                      |

Use the point/history and applicable interval/report UI for the January 15 local day. Sum **kWh**, and take the maximum of **kW**. Do not sum demand or infer success from the task log. Record actual results, times, point IDs and evidence in the run record. Report partial success at the failed boundary if collection, publication, storage or rendering does not complete.

## 5. Extend the same implementation

1. Add Science Center Electric Main (MV90 MDEF) under System / Northlake University / Northlake Main Campus / Science Center with its two points, Delivered kWh 15m and Demand kW 15m.
2. Configure and calculate the campus aggregates Campus Interval Electric kWh Total and Campus Coincident Electric kW directly under Northlake Main Campus. They cover the four interval-metered campus buildings, so Science Center Electric Main, Student Center Electric Main, Lakeview Hall Electric Main and Recreation Center Electric Main must all exist with their points first. Coincident demand is the maximum of the interval-by-interval sum, not the sum of separate meter peaks.
3. Add the KSAC Sacramento Executive Airport station meter under System / Northlake University / Weather Reference and its HDD/CDD relationships through the index UI, confirming base temperature, method and station assignments (KSAC for every building in all three companies).
4. Add the Student Center Kitchen Gas MVRS and Cedar Row B Water Master Neptune register/usage pairs (Register Reading with Route Usage Therms, and Register Reading with Route Usage kgal). Create the accumulator through the UI and capture its related usage point rather than creating a disconnected duplicate. Replace all placeholder handheld IDs, including meter and register IDs.
5. Work through the remaining gateway format cases in the coverage list. Use separate restores for alternative formats that share a measurement.

Rate modeling, bills, AP/GL, user permissions, tenants and broader reporting continue through the existing phase plans after the measurement foundation. The chiller status point, activity index, baselines and other feeds remain explicitly pending until their own input paths and results have been exercised.

## **6. Promote a restore baseline**

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Save the database backup only after the applicable acceptance checks pass. Keep the application version, packet/source revision, fixture hashes, actual-ID mapping, instance configuration and observed results beside it. Gateway file directories and external services are not reproduced by a database backup alone.

Restore once into a disposable test instance, reconnect its intended sources and verify the same saved data through the UI before calling it the Northlake baseline. Keep the configured-empty checkpoint as well as the accepted-data checkpoint so replay and backfill tests can start cleanly.